Q7. Descriptive Statistics

Original Aim: To calculate descriptive statistics for the numerical attributes of the Iris dataset. Find the mean, median, sample variance and standard deviation for SepalLengthCm and PetalLengthCm. Display the results in a clear table and briefly compare the spread of the two attributes.

Aim

Calculate mean, median, sample variance and sample standard deviation for SepalLengthCm and PetalLengthCm, then compare their spread.

Requirements

Python, Pandas and scikit-learn. Reference: [D7].

Essential theory

Mean is the arithmetic average; median is the middle ordered value. Sample variance sums squared deviations from the mean and divides by n-1. Sample standard deviation is its square root. Variance is in squared centimetres; standard deviation is in centimetres, making it easier to compare with the original measurements.

Step-by-step procedure

1. Select only the two assigned numerical columns.
2. Calculate mean and median.
3. Use var(ddof=1) and std(ddof=1) to request sample definitions explicitly.
4. Combine the four summaries into one table and round for display.
5. Compare spread using standard deviations and variances, noting that both columns share the same physical unit.

Complete Python program

```
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

selected = df[["SepalLengthCm", "PetalLengthCm"]]
stats_table = pd.DataFrame({
    "Mean": selected.mean(),
    "Median": selected.median(),
    "Sample variance": selected.var(ddof=1),
    "Sample std": selected.std(ddof=1)
})
print(stats_table.round(6).to_string())
```

Code explanation

Each reduction operates columnwise by default. The dictionary passed to DataFrame names the output statistics, while each Series keeps its original feature index; Pandas aligns them into rows. `round(6)` affects only the displayed table because the underlying stats_table remains unrounded. ddof=1 means one degree of freedom is subtracted from the denominator.

Verified output

	Mean	Median	Sample variance	Sample std
SepalLengthCm	5.843333	5.80	0.685694	0.828066
PetalLengthCm	3.758000	4.35	3.116278	1.765298

Observation

PetalLengthCm has standard deviation about 1.7653 cm and variance 3.1163 cm^2, compared with 0.8281 cm and 0.6857 cm^2 for SepalLengthCm. Petal length therefore shows greater absolute spread in this pooled dataset. The petal mean 3.758 differs from its median 4.35, reflecting the distribution's grouped structure; that difference alone should not be used as a complete shape diagnosis.

Result

All four requested sample summaries were tabulated and showed greater variability in petal length than sepal length. No population ddof=0 result was substituted for the requested sample result.

Common errors and troubleshooting

- NumPy gives a slightly smaller standard deviation: its default is ddof=0; specify ddof=1 for sample calculations.
- Variance is described as centimetres: variance uses cm^2; its square root returns cm.
- Species enters the calculation: select the assigned measurement columns explicitly.

Viva questions and answers

- Why divide sample variance by n-1? Estimating the mean uses one degree of freedom; this yields the usual unbiased sample-variance estimator under standard assumptions.
- What is the relation between variance and standard deviation? Standard deviation is the positive square root of variance.
- Which measure is more robust to extremes, mean or median? The median.
- Does larger standard deviation always mean a worse feature? No; spread is descriptive, not a direct measure of predictive usefulness.